

Saqib Ahmad

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building **Java/Spring Boot microservices** and REST APIs using OOP principles in enterprise environments. Experienced with **Spring Security, OAuth 2.0, JPA/Hibernate, Kafka, and Redis**, along with full-stack development using **React, Node.js, Express.js, and MongoDB**. Contributed to enterprise platforms processing **1M+ Kafka events/day**, with experience in API optimization and production debugging. Built an **AI-powered GitHub code assistant** using **Spring AI, OpenAI API, RAG, semantic search, and pgvector**.

WORK EXPERIENCE

Associate Software Engineer | Tech Mahindra

Sep 2024- Aug 2026 | Bangalore

- Developed **Java/Spring Boot microservices** and **OAuth 2.0-secured REST APIs** for Optus's **5G Standalone (SA)** demand, handling **25K+ requests/day** at **sub-200ms API response times**.
- Improved API performance by **25%** using **Redis** caching and indexed plan-policy lookups, eliminating full-table scans; implemented **Kafka-based** asynchronous processing to decouple provisioning microservices.
- Reduced recurring production defects by **40%** by tracing and fixing root causes via **SLF4J/Logback** structured logging; implemented **JPA/Hibernate** persistence with centralized validation and exception handling.
- Architected RESTful API endpoints using Node.js/Express.js secured with JWT-based authentication/authorization, and built ReactJS UI modules for enquiry, booking, and lead-capture workflow on Project Drona.
- Modeled MongoDB schemas and implemented CRUD endpoints with a middleware validation pipeline and error handling; debugged production issues in an Agile workflow.
- Executed 250+ functional test cases across E2E service workflows using SoapUI, validating request/response behavior and cross-service integration, and logging 20+ defects in HP ALM pre-production.

PROJECTS

DevPilot 🚀

Java, Spring AI, Spring Boot, RAG, GenAI, PostgreSQL/pgvector

- Built a **RAG** pipeline for DevPilot, a **Java/Spring Boot LLM** assistant, embedding GitHub source via **text-embedding-3-small/Spring AI** into **PostgreSQL/pgvector (1536-dim)** with **HNSW/cosine** similarity.
- Enabled repository-scoped **semantic search**, retrieving **top-8 chunks** via metadata-filtered vector similarity, and streamed **GPT-4o-mini** responses token-by-token via **SSE** with file/line citations for grounded answers.
- Automated non-blocking repository indexing via **Spring @Async** and a **thread-pool executor**, batching vector inserts (**32 docs**) with **50ms GitHub rate limiter**; secured via **OAuth 2.0** and **AES-encrypted** token storage.

Deep Packet Inspection (DPI) Engine 🛡️

Java, Multithreading, Networking, PCAP

- Engineered a byte-level packet inspection pipeline for Ethernet/IPv4/TCP/UDP protocol dissection and TLS ClientHello parsing, extracting SNI metadata for application-layer classification of encrypted HTTPS traffic (YouTube, Facebook, GitHub, etc.) without decrypting payloads.
- Reduced PCAP processing time by 30% versus a 1-LB/1-FP baseline on a 5.47 MB capture by parallelizing DPI classification across 2 Load Balancers and 4-FP workers pipeline with consistent-hashing-based flow routing.

TECHNICAL SKILLS

- Languages** : C, C++, Java, Python, JavaScript, TypeScript, SQL, HTML/CSS
- Backend & Security** : Spring, Spring Boot, Spring Security, Spring AI, Microservices, Node.js, Express.js, REST APIs, JPA/Hibernate, ORM, OAuth 2.0, JWT, Load Balancer, Relational/Non-Relational Databases
- AI & RAG** : OpenAI API, LLM, RAG, Embeddings, Semantic/Vector Search, pgvector, HNSW, Server-Sent Events
- Frontend & Data** : React/ReactJs, Next.js, Redux, Tailwind, MongoDB, Mongoose, MySQL, PostgreSQL, Redis
- Messaging & Testing** : Kafka, Multithreading, Swagger/OpenAPI, JUnit/Mockito, Postman, SoapUI, JIRA, Alm
- Tools** : Git, GitHub, Maven, ELK(Elasticsearch/Logstash/Kibana), SLF4J/Logback, SMTP, Docker
- Core CS** : Data Structures and Algorithms, Object-Oriented Programming, Java Streams & Java concurrency, OS

EDUCATION

Bundelkhand Institute of Engineering & Technology

Bachelor of Technology in Computer Science & Engineering, CGPA:- **7.96**

Jhansi, UP

2020 - 2024

ACHIEVEMENTS

- Ranked **443** globally in **Google Kickstart Round H** 🚀
- 3-Star Competitive Programmer** at **CodeChef** 🚀
- Microsoft Certified: Azure Data Fundamentals** 🚀